

Theorem (Bernoulli's inequality). *Let $x \in \mathbb{R}$ with $x \geq -1$ and let $n \in \mathbb{Z}_{\geq 0}$. Then*

$$(1+x)^n \geq 1+nx.$$

Equality holds exactly in one of the following cases

- $n = 0$ (any admissible x);
- $x = 0$ (any $n \geq 0$);
- $x = -1$ and $n = 1$.

Proof. **1. Degenerate cases.**

- $n = 0$. $(1+x)^0 = 1$ and $1+0 \cdot x = 1$, so equality holds.
- $x = -1$. Then $(1+x)^n = 0$ and the right-hand side equals $1-n$. Since $n \geq 0$, $0 \geq 1-n$ is equivalent to $n \geq 1$, so the inequality is true for all $n \geq 1$. Equality $0 = 1-n$ occurs only when $n = 1$.
- $x = 0$. Both sides equal 1, which will also appear in the general equality analysis.

Henceforth we assume $n \geq 1$ and $x > -1$ (the case $x = -1$ already settled).

2. Proof for $x \geq 0$ (binomial expansion).

The Binomial Theorem gives, for any integer $n \geq 0$,

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k. \quad (1)$$

Separating the first two terms,

$$(1+x)^n = 1 + nx + \underbrace{\sum_{k=2}^n \binom{n}{k} x^k}_S. \quad (2)$$

When $x \geq 0$ each binomial coefficient $\binom{n}{k} > 0$ and each power $x^k \geq 0$, hence $S \geq 0$. Consequently

$$(1+x)^n \geq 1+nx.$$

If equality holds then $S = 0$. Because every term in S is non-negative and the coefficients are positive, we must have $x^k = 0$ for all $k \geq 2$, i.e. $x = 0$. This yields the equality case $x = 0$ (with any n).

3. Proof for the whole interval $[-1, \infty)$ (induction).

Base step $n = 0$. Already covered in the degenerate cases.

Inductive hypothesis. Assume for a fixed $n \geq 0$

$$(1+x)^n \geq 1+nx \quad \text{for every } x \geq -1. \quad (\text{H}_n)$$

Inductive step. Since $x \geq -1$ we have $1 + x \geq 0$, so multiplying (??) by the non-negative factor $1 + x$ preserves the inequality:

$$(1 + x)^{n+1} \geq (1 + nx)(1 + x). \quad (3)$$

Expanding the right-hand side,

$$(1 + nx)(1 + x) = 1 + (n + 1)x + nx^2. \quad (4)$$

Substituting (??) into (??) gives

$$(1 + x)^{n+1} \geq 1 + (n + 1)x + nx^2. \quad (5)$$

Because $n \geq 0$ and $x^2 \geq 0$, the term nx^2 is non-negative. Dropping this non-negative term yields

$$(1 + x)^{n+1} \geq 1 + (n + 1)x,$$

which is the desired inequality for $n + 1$.

Equality analysis in the inductive step. In (??) the extra term nx^2 vanishes precisely when $n = 0$ or $x = 0$. If $n \geq 1$ and $x \neq 0$, the term is positive, making the inequality strict. The case $x = -1$ has been handled separately; equality then occurs only for $n = 1$.

Thus, by induction, Bernoulli's inequality holds for every integer $n \geq 0$ and every real $x \geq -1$.

4. Equality condition (combined).

From (??) we have

$$(1 + x)^n = 1 + nx + S, \quad S := \sum_{k=2}^n \binom{n}{k} x^k \geq 0.$$

Equality $(1 + x)^n = 1 + nx$ is equivalent to $S = 0$. Since each coefficient $\binom{n}{k} > 0$, each term $\binom{n}{k} x^k$ must be zero. Hence either $x = 0$ (forcing all terms with $k \geq 2$ to vanish) or the sum is empty, i.e. $n \leq 1$.

When $n = 0$ the statement reduces to $1 = 1$ for any admissible x . When $n = 1$ the inequality reads $(1 + x) \geq 1 + x$, which is an equality for every x ; in particular for the endpoint $x = -1$ we obtain the extra equality case $x = -1$, $n = 1$.

Collecting all possibilities, equality holds exactly in the three cases listed at the beginning of the theorem.

□